Final Review Spring 2020

This was done in my Spring 2020 final review session! You are welcome to use it if it's helpful to you.

- How will the questions about writing and reading files be formatted?
  - For example:
  - Given a CSV file that looks like this:

    2.5,3.3,5.0
    3.3,2.2
    1.5,6.7,3.4
  - How would you traverse over it to get a sum of all numbers in the file?

```
def csv_sum(fname):
    file = open(fname)
    total = 0
    for line in file:
        line = line.strip().split(',')
        # What is our line now?
        ## it's a list!
        for item in line:
            total += float(item)
    file.close()
    return total
```

- For the rectangle class, how come the area and perimeter went before __repr__?
  - They didn't have to!
- When do we use range()?
  - range() is what we call a "generator" - it generates a set of numbers between 0 (or whatever minimum you provide) through the maximum value you provide, optionally by a certain factor
  - We use it to traverse over sequences (lists, strings) by index!
    - `for i in range(len(lst)):`
    - `for i in range(1, 11, 2) # by = 2`
- What do we need to know about files?
  - See above!
  - The things we learned about files:
    - How to open a file in read mode
    - How to open a file in write mode
    - How to traverse a file by line
    - How to accumulate values in python from values read in files
- How do we sort keys of a dictionary in sorted order?
  - `sorted(dict)`
    - Returns a list of sorted keys
  - Use to traverse over a dictionary with the keys in sorted order:
    - `for key in sorted(dict):`

      …